



Fig. 5: Animal RFID glass tag LF 125kHz RFID reader (Credit: www.rfid-animaltags.com)

Medical Device ID		Boston Scientific	
Patient: Patient Name			
Physician: Following Physician/Clinic Name			
Physician Telephone: Following Phone			
MFG	Product	Model/Serial	Implant Dt
Boston Sci	CRT-D	1234 968785	12-OCT-2015
Boston Sci	Lead	7654 123456	12-OCT-2015
Boston Sci	Lead	6789 234567	12-OCT-2015
Boston Sci	Lead	4321 345678	12-OCT-2015
Implanted Cardiac Rhythm Management Patient Contact physician for medical questions or emergency www.bostonscientific.com			
For Patients 866.454.3255 (USA) 001.651.582.4000 (Outside USA) www.lifebeatonline.com		For Security Personnel Magnetic security wands may adversely affect device function. Do not place wand over device.	
For Medical Personnel 1.800.227.3422 or 651.582.4000			
MRI - For questions regarding MRI device compatibility, see the Boston Scientific website at www.bostonscientific.com/mageready			

Fig. 6: In an emergency, a medical device identification card alerts medical personnel that the patient has an implanted device (Credit: www.bostonscientific.com)

with the outside world must conform to IEEE human tissue exposure standards, namely, IEEE 1999 and IEEE 2002.

Communication technology for human implant devices. These devices use communication through electromagnetic waves to exchange data between an interrogator (also called reader) and transponder for identification and tracking purposes. The transponder generally consists of an IC for storing and processing data signals, and an antenna for receiving and transmitting signals.

The device can be fabricated at bio-molecular scale that interrogates its local chemical environment (sensing) and/or controls the biological machinery inside a single living cell. Remote monitoring at such a small scale provides the selectivity requirement, that is, local control of bio-molecular machinery without damaging cells or affecting biological events in the surrounding medium.

The technology allows real-time remote monitoring of fully-autonomous implants in the human body. Read range of communication for these implants is about 10cm. Such short-read range is advantageous for meeting security or

been successfully demonstrated by scientists. They have done so by implanting a nano-robot in embedded smart sensors in the human body for tele-operated surgical procedures or early disease diagnosis.

Undesirable migration of the device inside the body needs to be controlled. Secure and reliable wireless transmission of personal or confidential data recorded by the device inside the body needs to be established.

Total confidentiality and high reliability are important criteria to implement such wireless devices. For safety to antennae implanted in human tissue, electromagnetic energy radiated or re-radiated by an implant device for wirelessly communicating

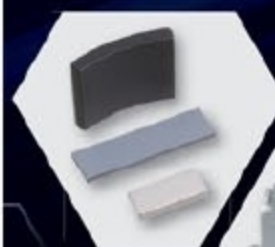
Attracting Tomorrow



TDK Technology Advancing power solutions.

Rare-earth magnets
with high magnetic
field strength for wind
power generators

**Varistors and
surge arresters**
with high surge
capability



**EMC filters and
sine-wave filters**
for currents up to 8 kA

**Aluminum electrolytic
and film capacitors**
for high ripple currents



product.tdk.com/en/industrial